

Static Gesture Classification — Run Report

Generated 2026-05-13 21:44:19

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	16 · 559 trials total
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Split / seed / CV	random_state=42 · CV: LOUO (16 folds, every user held out once)
Sample counts	train (post-aug)=559 · LOUO-evaluated samples=559

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — LOUO cross-validation

Algorithm	CV mean (LOUO, 16 folds)	CV std	CV min	CV max
Random Forest	0.9306	0.0721	0.7714	1.0000

Per-class CV F1 (across all LOUO folds)

Class	Random Forest	Support
Double_Closed_Fist	0.936	83
Double_Nothing	0.766	75
Double_Okay	0.994	80
Double_Open_Palm	0.982	80
Double_Phone	0.928	80
Double_Pistol	0.902	81
Double_Spiderman	0.982	80
macro avg	0.927	559

Per-user accuracy (each user as held-out fold)

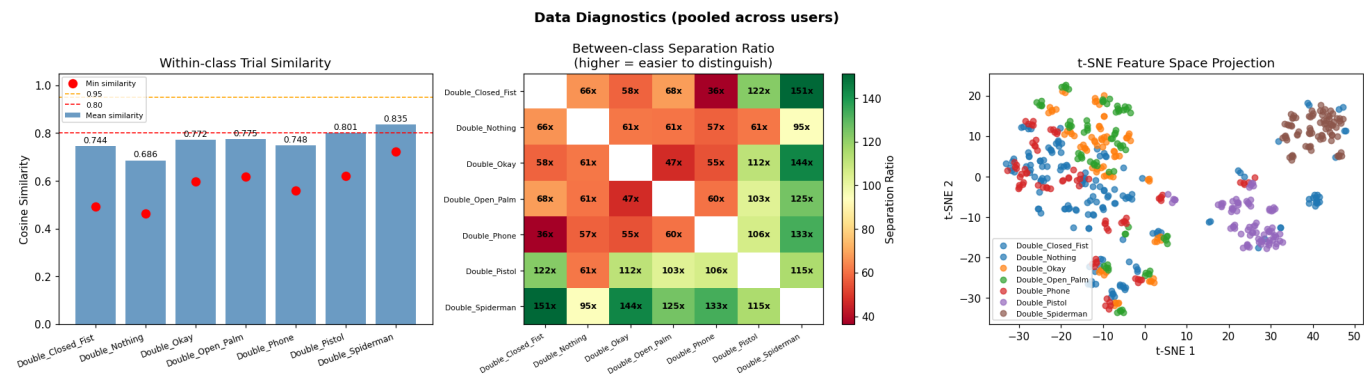
User	n trials	Random Forest
1_Alan	35	1.000
2_Alex	35	0.771
3_Anghad	35	0.886
4_Daniel	35	0.943

5_Georgia	35	1.000
6_Harry	35	0.857
7_Henry	35	0.829
8_Jestin	36	1.000
9_Joselyn	30	1.000
10_Josh	35	0.857
11_Mansh	36	1.000
12_Marcus	35	1.000
13_Nat	37	0.919
14_StephenV2	35	0.971
16_Bella	35	0.886
17_Raquel	35	0.971
Mean (across users)		0.931
Std (across users)		0.072

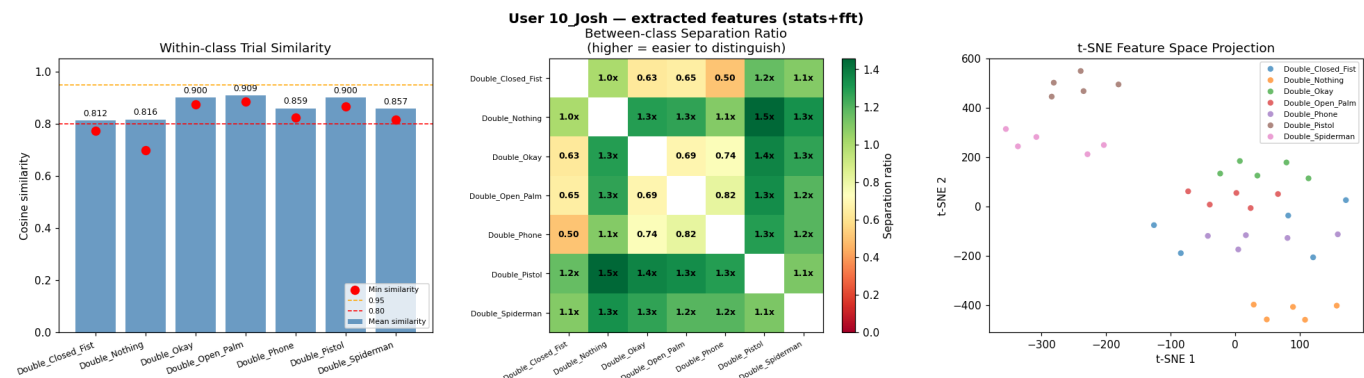
Appendix — Data Diagnostics

17 figures: pooled across users + one per detected user (extracted-feature views).

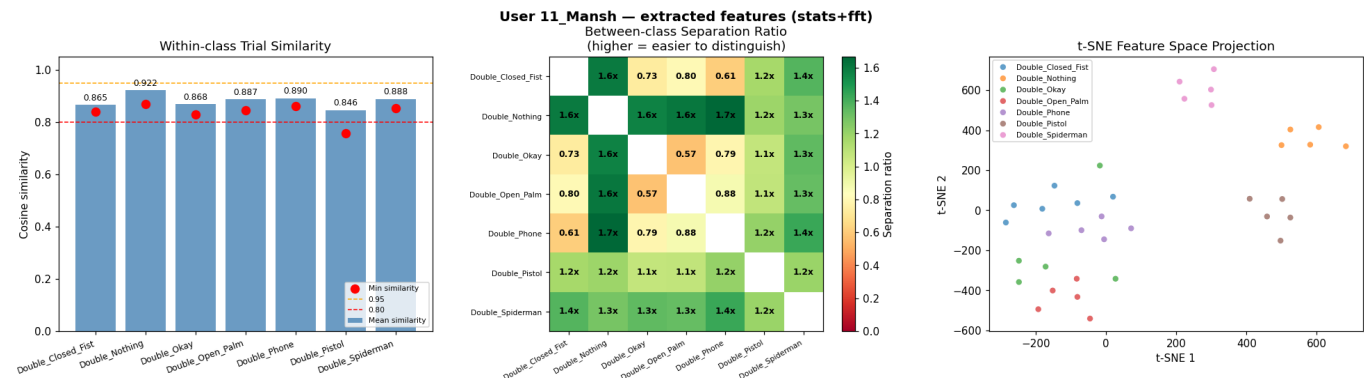
Pooled diagnostics — extracted features (across all users)



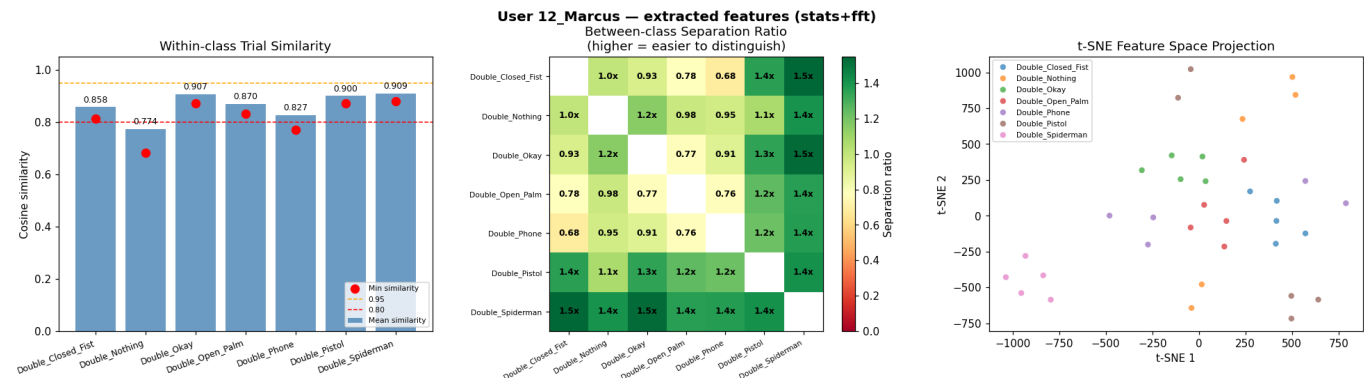
User 10_Josh — extracted features (stats+fft)



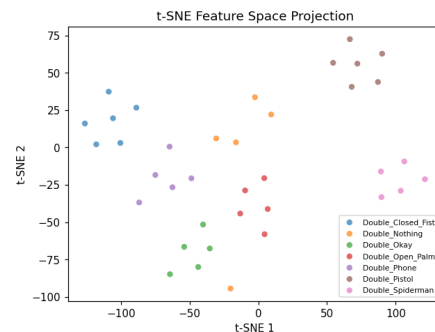
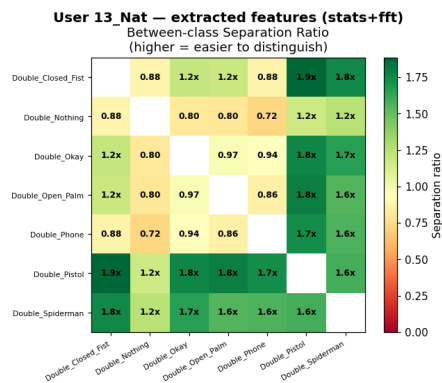
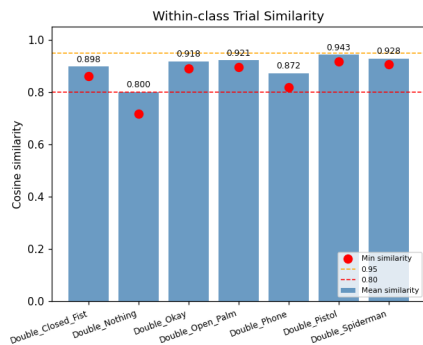
User 11_Mansh — extracted features (stats+fft)



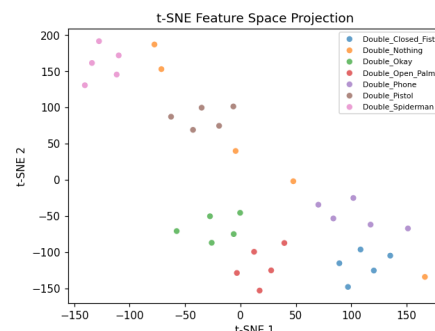
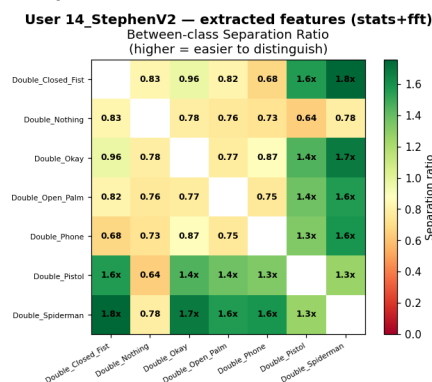
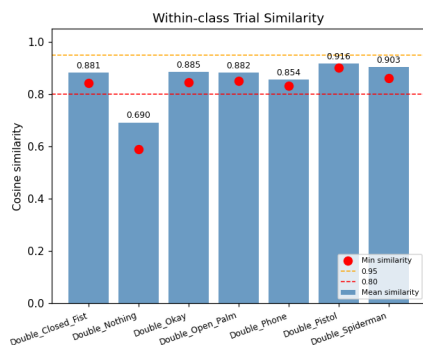
User 12_Marcus — extracted features (stats+fft)



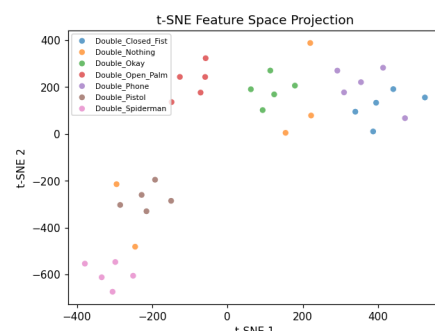
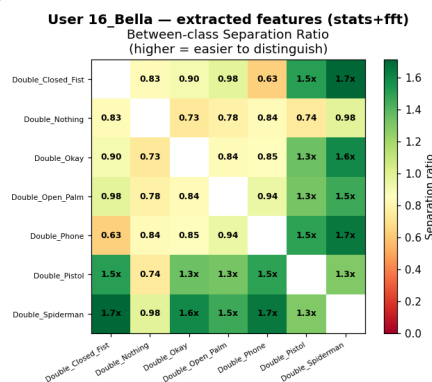
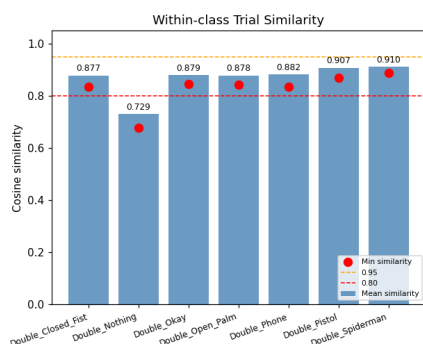
User 13_Nat — extracted features (stats+fft)



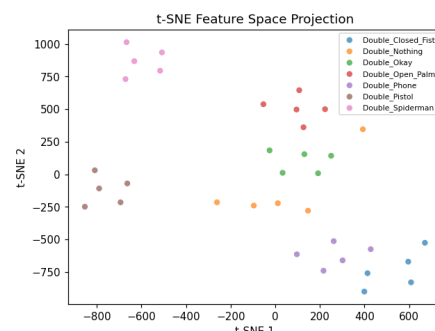
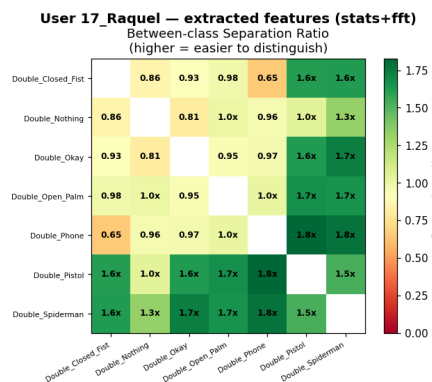
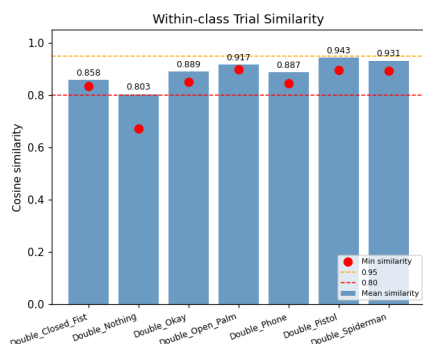
User 14_StephenV2 — extracted features (stats+fft)



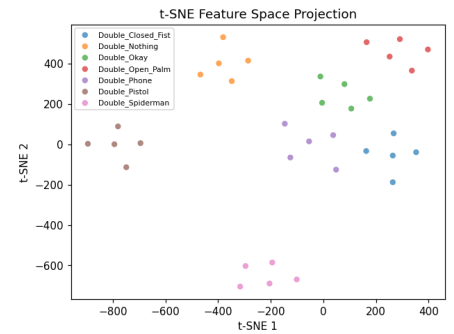
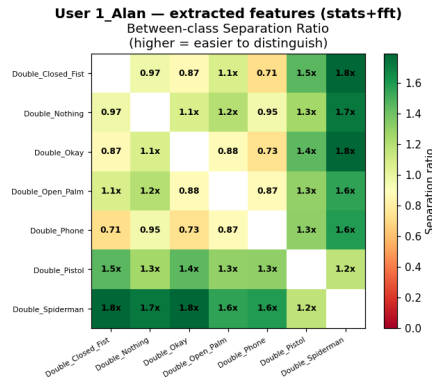
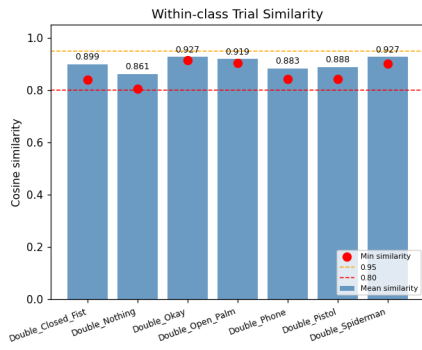
User 16_Bella — extracted features (stats+fft)



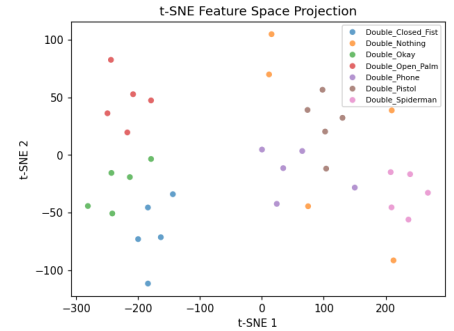
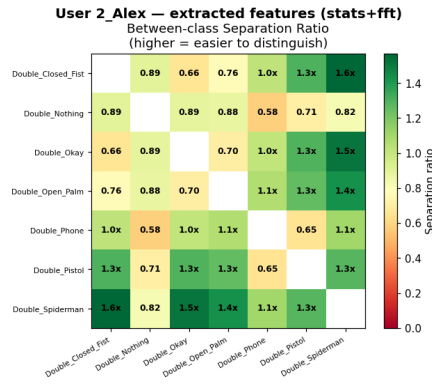
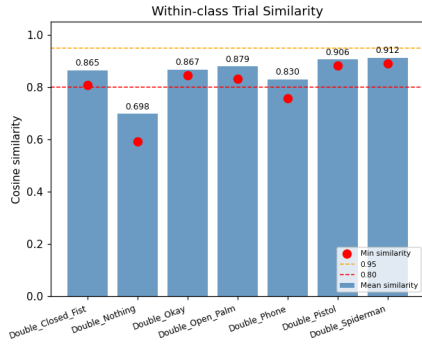
User 17_Raquel — extracted features (stats+fft)



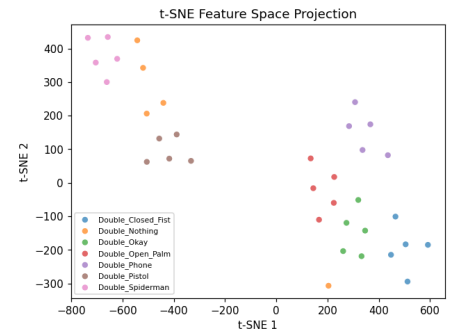
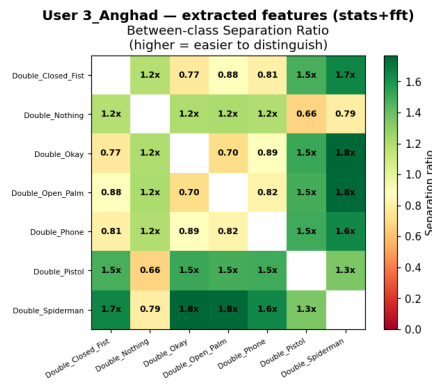
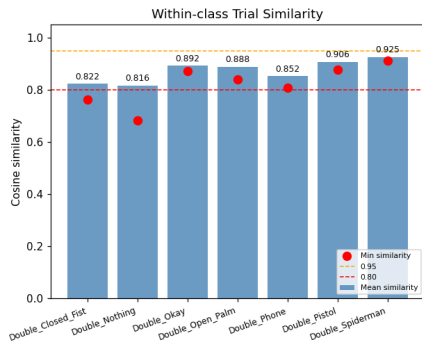
User 1_Alán — extracted features (stats+fft)



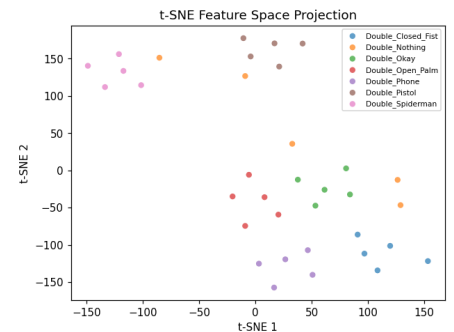
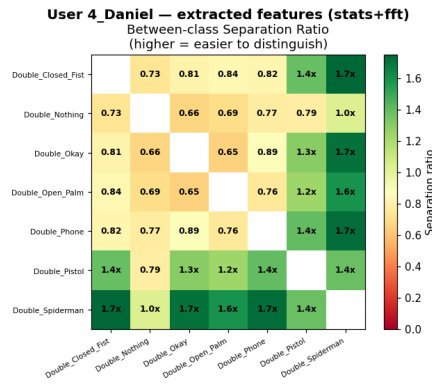
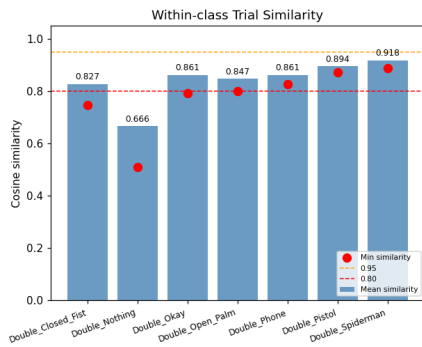
User 2_Alex — extracted features (stats+fft)



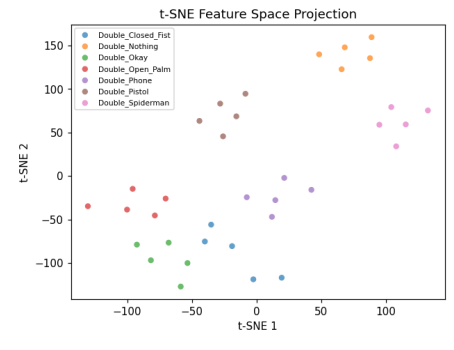
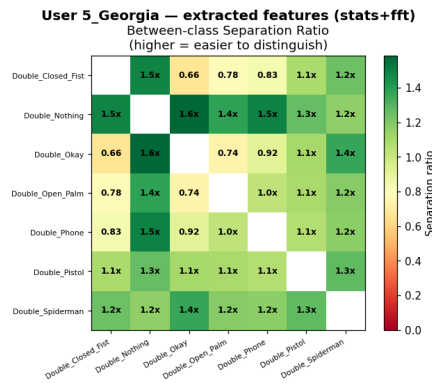
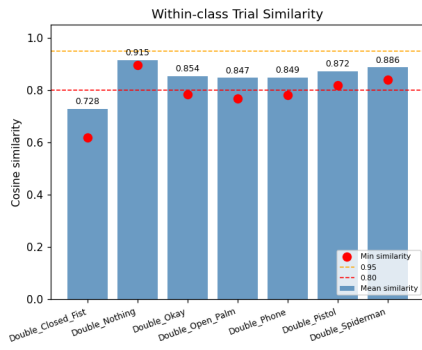
User 3_Anghad — extracted features (stats+fft)



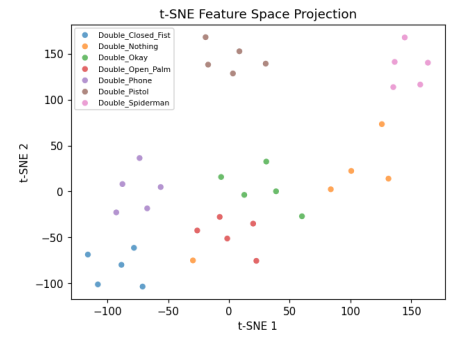
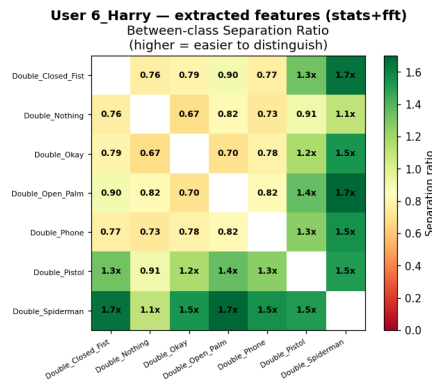
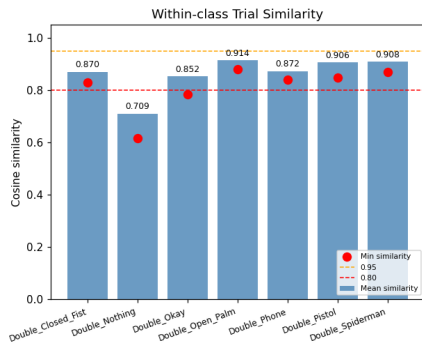
User 4_Daniel — extracted features (stats+fft)



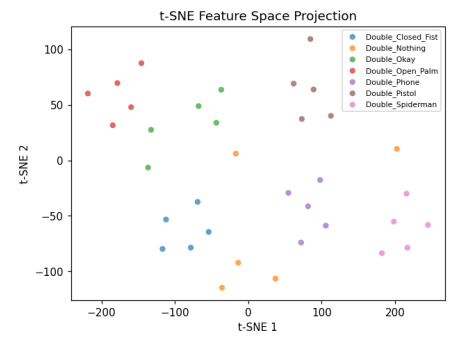
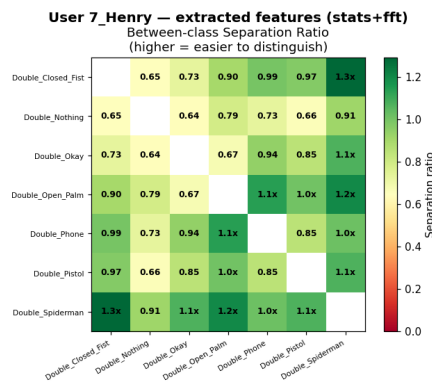
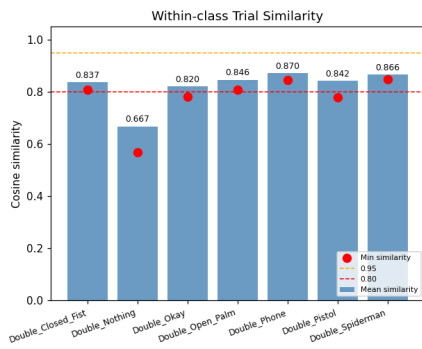
User 5_Georgia — extracted features (stats+fft)



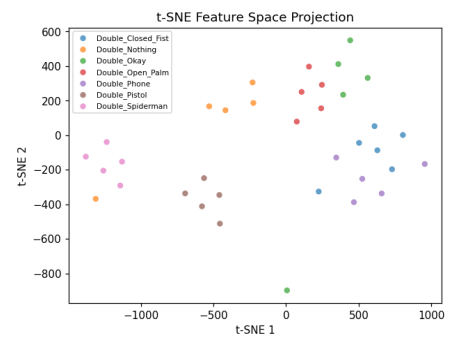
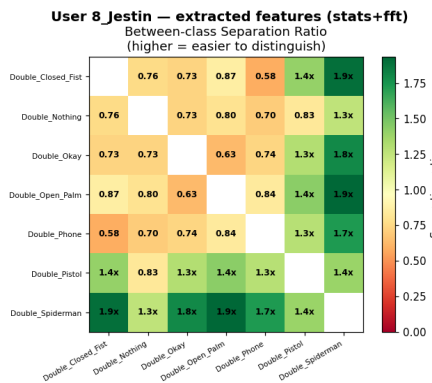
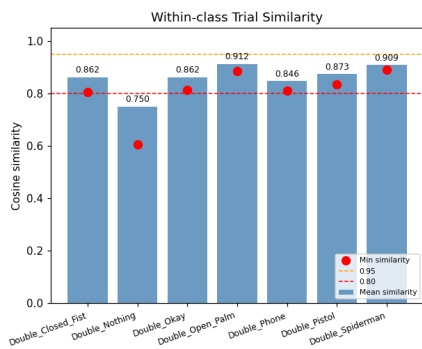
User 6_Harry — extracted features (stats+fft)



User 7_Henry — extracted features (stats+fft)



User 8_Jestin — extracted features (stats+fft)



User 9_Joselyn — extracted features (stats+fft)

